

Automatic weak-star continuity answers two questions in arXiv:0012032

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered`: automatic weak-star continuity of left multipliers of dual operator spaces, and the equivalent module-action question.

The exact questions

Let X be a dual operator space. A left multiplier of X is a linear map $u : X \rightarrow X$ which, after a completely isometric realization $\sigma : X \rightarrow A$ in a C^* -algebra, is implemented by left multiplication: $\sigma(ux) = a\sigma(x)$. The target paper proves weak-star continuity for left *adjointable* multipliers but states that it does not know whether every $u \in \mathcal{M}_l(X)$ is weak-star continuous. It therefore calls X *left normal* when all of its left multipliers have this property [1].

Later in the same paper, the authors ask the equivalent module question. If $m : B \times X \rightarrow X$ is a left operator-module action of a dual operator algebra and m is weak-star continuous in its first variable, must it also be weak-star continuous in the second? Their corollary proves that a counterexample exists if and only if some dual operator space has a left multiplier which is not weak-star continuous [1].

The later theorem

Blecher and Magajna subsequently proved the exact automatic-continuity statement:

Every left multiplier of a dual operator space is weak-star continuous.

This is Theorem 4.1 of arXiv:math/0407220, later published in the *Journal of Functional Analysis* [2]. Their proof uses the canonical completely contractive projection $q : X^{**} \rightarrow X$ associated with the given predual. For $u \in \mathcal{M}_l(X)$ they establish

$$q(u^{**}(\eta)) = u(q(\eta)) \quad (\eta \in X^{**}),$$

which is precisely the bidual criterion forcing u to be weak-star continuous. The paper explicitly records that the earlier “left normal” hypothesis is consequently automatic and can be removed.

The target’s own equivalence now settles its second question immediately. Indeed, the theorem rules out a non-weak-star-continuous left multiplier on any dual operator space. Hence a left module action as above which is weak-star continuous in the algebra variable cannot fail weak-star continuity in the space variable. Such actions are separately weak-star continuous.

Conclusion and scope

The selected pair of questions from arXiv:0012032 received a complete affirmative answer in arXiv:math/0407220. This classification is deliberately narrow. The target also asks two different representation questions: whether a unital operator algebra which is merely a dual Banach space has an isometric weak-star-homeomorphic concrete representation, and whether a unital function algebra with a Banach predual can be represented as a weak-star-closed subalgebra of an L^∞ space. The multiplier theorem does not answer either representation question, and this note makes no claim about their present status.

References

- [1] D. P. Blecher and B. Magajna, *Multipliers and dual operator algebras*, arXiv:math/0012032, <https://arxiv.org/abs/math/0012032>.
- [2] D. P. Blecher and B. Magajna, *Duality and operator algebras: automatic weak* continuity and applications*, J. Funct. Anal. **224** (2005), 386–407, <https://doi.org/10.1016/j.jfa.2004.10.013>; arXiv:math/0407220, <https://arxiv.org/abs/math/0407220>.